

Overview of Ecosystem Service: Coastal Carbon Storage

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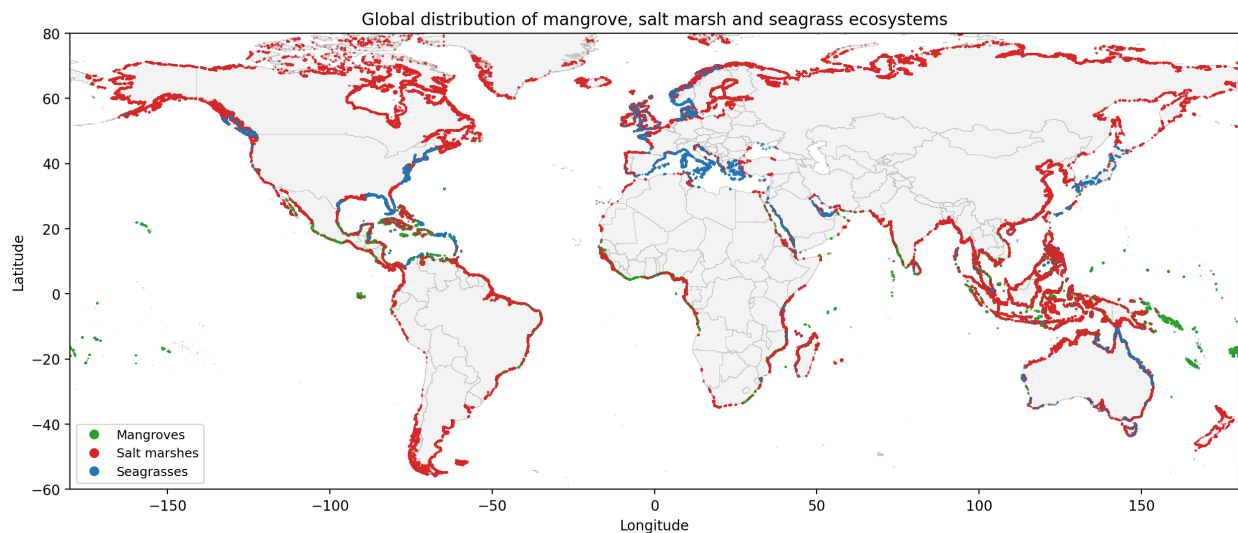
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1 Introduction

Carbon storage in mangroves, salt marshes and seagrass meadows is an essential coastal “blue carbon” ecosystem service for climate change mitigation (McLeod et al. 2011; Howard et al. 2014; IPCC 2014; Macreadie et al. 2021) (Figure 1). Here we provide a comprehensive, global, country-level economic assessment of the carbon stock held in these three BCE types within the Gross Ecosystem Product (GEP) framework. We focus on the standing stock of organic carbon contained within each coastal country’s marine Exclusive Economic Zone (EEZ), following the country-level blue carbon natural-capital accounting framework of Bertram et al. (Bertram et al. 2021), and value that stock through the rental social cost of carbon.

Figure 1: Global distribution of mangrove, salt marsh and seagrass ecosystems



Notes: This figure is plotted from the Global Mangrove Watch v3 (Bunting et al. 2018), the GLC_FCS30D salt marsh layer (Zhang et al. 2024) and the UNEP-WCMC Global Distribution of Seagrasses v7.1 (UNEP-WCMC and Short 2021); the ecosystem-trio framing follows Macreadie et al. (Macreadie et al. 2021).

2 Data and Methods

The spatial domain of this analysis is the marine Exclusive Economic Zone (EEZ) of each coastal country, which we treat as the coastal area attributed to that country, following the country-level blue-carbon natural-capital accounting framework of Bertram et al. (Bertram et al. 2021). Restricting the domain to EEZs delineates the coastal-water envelope where mangrove, salt marsh and seagrass ecosystems occur, and, by construction, ensures that the coastal-carbon accounts reported here do not spatially overlap with the terrestrial-carbon service, which is reported on land within the standard terrestrial country boundaries.

2.1 Data sources

Mangrove extent comes from the Global Mangrove Watch v3 vector product for 2019 (Bunting et al. 2018), which integrates ALOS PALSAR and Landsat observations to map mangrove forests at 30 m globally. Salt marsh extent is taken from a pre-processed vector layer extracted from the Zhang et al. GLC_FCS30D global land-cover product for 2019 (Zhang et al. 2024). Seagrass extent comes from the UNEP-WCMC Global Distribution of Seagrasses v7.1 polygon product (UNEP-WCMC and Short 2021) (293,147 polygons globally), released in 2021 as a multi-year compilation of field observations spanning the late nineteenth century to 2018. No global wall-to-wall seagrass map exists for the 2019 base year used here, but seagrass meadows are slow-changing ecosystems whose extent persists on decadal timescales, so the WCMC v7.1 compilation is treated as the best available 2019 baseline. The area totals it produces sit within the high- to low-confidence global extent range of 160,000 to 267,000 km² synthesised by McKenzie et al. (McKenzie et al. 2020) from the same underlying field record, which is the extent layer adopted by the most recent global seagrass blue-carbon synthesis (Gomis et al. 2025); in other words, WCMC v7.1 is the polygon-level realisation of the same observational record that McKenzie and Gomis use in tabular form.

Mangrove soil organic carbon is taken from the Sanderman et al. random-forest map of mangrove soil organic carbon at 30 m, top 1 m depth (Sanderman et al. 2018). We use the median-typology product for the 2019 to 2020 epoch; the file is reprojected once onto the 10 arc-second grid using average resampling and cached on disk for subsequent runs. Salt marsh soil carbon is taken from a latitude step function summarising the Maxwell et al. MarSOC synthesis (Maxwell et al. 2024), with values rising from ~180 Mg C/ha in the tropics to ~350 Mg C/ha at high latitudes. Seagrass soil carbon is set to a Fourqurean et al. (Fourqurean et al. 2012) global mean of 22 Mg C/ha for the top 20 cm, scaled to ~70 Mg C/ha for the top 1 m to align with the soil-depth convention used for mangrove and salt marsh; the scalar assumes a roughly linear depth profile and is a known approximation that the source dataset does not directly support.

2.2 Quantifying the blue carbon stock

The country-level storage value is built from a quantity step and a pricing step. The quantity step, described in this subsection, estimates the standing organic carbon stock in each coastal country's EEZ; the pricing step, described in the following subsection, converts that stock into an annualised monetary value. For each ecosystem the per-pixel (or per-polygon, for seagrass) carbon stock is the sum of above-ground biomass carbon, below-ground biomass carbon and soil organic carbon to 1 m depth, each term drawn from the published global product appropriate to that ecosystem.

2.2.1 Mangroves

Above-ground biomass density is computed from the Hamilton and Friess (Hamilton and Friess 2018) combined latitude regression, $AGB = \max(0, -6.4305|\text{lat}| + 271.747)$ in tonnes per hectare, which averages the earlier Twilley et al. (Twilley, Chen, and Hargis 1992) and Saenger and Snedaker (Saenger and Snedaker 1993) equations and explains roughly four times more variance than any single prior latitude-only model. AGB is converted to carbon at 0.48 following Hamilton and Friess (Hamilton and Friess 2018).

Below-ground biomass is allocated from AGB using the IPCC (IPCC 2014) Wetlands Supplement Table 4.6 climate-zone ratios, in line with the recent Wang et al. mangrove storage map (Wang et al. 2025). Tropical wet pixels ($|\text{lat}| < 23.5^\circ$ and mean annual precipitation greater than 2,000 mm) use a BGB:AGB ratio of 0.49; tropical dry pixels (same latitude range, lower precipitation) use 0.29; and subtropical pixels ($|\text{lat}| \geq 23.5^\circ$) use 0.96. When no precipitation raster is configured, all tropical pixels default to the wet-tropical ratio, which is the dominant condition for global mangroves but slightly overestimates BGB along the Arabian and Australian dry-tropical coasts. BGB is converted to carbon at 0.39 following Howard et al. (Howard et al. 2014).

Soil organic carbon is read directly from the Sanderman map at the pixel scale, with no further parameterisation. As a fallback when the Sanderman raster is not available locally, the pipeline reverts to a latitude step function (decreasing from 400 Mg C/ha at the equator to 250 Mg C/ha by 25° latitude) that approximates the Sanderman global mean. Total mangrove ecosystem carbon at a tropical wet equatorial pixel with Sanderman SOC = 400 Mg C/ha is therefore approximately $130 + 52 + 400 = 582$ Mg C/ha, consistent with the upper end of the Macreadie et al. published range (Macreadie et al. 2021) for equatorial mangroves.

2.2.2 Salt marshes

Above-ground biomass for salt marshes is set to a Chmura et al. global median of 5 t/ha (Chmura et al. 2003), boosted by a factor of 1.2 in tropical pixels ($|\text{lat}| < 25^\circ$) to reflect higher productivity in warm, perennially flooded settings. The conservative 2.5 below-ground to above-ground ratio captures the marsh root systems' large allocation to below-ground biomass, which sits within the published 2:1 to 5:1 range. Biomass is converted to carbon at 0.45 (AGB) and 0.41 (BGB) following Howard et al. (Howard et al. 2014).

Soil organic carbon is read from a four-bin latitude step function calibrated against the Maxwell et al. MarSOC synthesis (Maxwell et al. 2024): 180 Mg C/ha in the tropics, rising through 220 Mg C/ha in the warm temperate zone and 250 Mg C/ha in the cool temperate zone, to 350 Mg C/ha in peat-rich boreal and arctic marshes. A native MarSOC raster is not currently wired into the pipeline; the step function captures the latitudinal first order but not the substantial within-zone variability documented in the source dataset.

A representative salt marsh pixel at 30°N with median productivity and 220 Mg C/ha soil carbon yields approximately $7 + 220 = 227$ Mg C/ha total, reflecting the marsh's near-total reliance on soil for storage.

2.2.3 Seagrasses

Above- and below-ground biomass carbon densities are assigned per polygon using the genus-specific means from Gomis, Strydom, Foster et al. (Gomis et al. 2025), who report total biomass carbon densities spanning roughly two orders of magnitude across the twelve genera in the global database: 9.78 Mg C/ha for *Posidonia* (the highest), ~ 3.70 Mg C/ha for other persistent genera (*Phyllospadix*, *Enhalus*, *Cymodocea*, *Thalassia*, *Amphibolis*, *Thalassodendron*), ~ 0.94 Mg C/ha for opportunistic genera (*Zostera*, *Syringodium*, *Halodule*), 0.33 Mg C/ha for *Ruppia* and 0.19 Mg C/ha for *Halophila*. Polygons with no genus identification (78 percent of the WCMC dataset) receive the Gomis 2025 global mean of 1.55 Mg C/ha. The total biomass density is split 30 percent above-ground and 70 percent below-ground in line with the seagrass-wide finding that root and rhizome

systems dominate biomass (Howard et al. 2014; Fourqurean et al. 2012). Polygons whose genus is a freshwater plant (*Trapa*, *Myriophyllum*, *Valisneria*, *Najas*, *Heteranthera*, *Stuckenia*, *Utricularia*) are zeroed out before aggregation.

Soil organic carbon is set to a constant 70 Mg C/ha for the top 1 m, derived by scaling the Fourqurean et al. (Fourqurean et al. 2012) top 20 cm value of 22 Mg C/ha. This is the same uniform constant for every polygon; spatial heterogeneity is currently not captured, and adding a native MarSOC-style raster for seagrass would be a natural future improvement. Per-polygon stock is the density times the polygon area in hectares, computed in the EPSG:6933 equal-area projection after intersection with the country EEZ polygons.

2.3 Pricing the carbon stock with the rental SCC

To estimate the value of carbon storage and sequestration, we use the rental social cost of carbon (SCC) (Parisa et al. 2022; Groom and Venmans 2023; Venmans, Rickels, and Groom 2025) as the pricing metric. Parisa et al. (Parisa et al. 2022) provide the theoretical rationale for quantifying the value of short-term carbon storage, which corresponds to carbon storage and sequestration. Following this approach, and within the Gross Ecosystem Product (GEP) framework, we estimate the annual rental value of carbon storage as

$$\text{rental SCC}(t) = P_C(t) - P_C(t+1)e^{-r} = P_C(t) - P_C(t)e^g e^{-r} = P_C(t)(1 - e^{-x}),$$

where $P_C(t)$ is the social cost of carbon in year t , $P_C(t+1)$ is the social cost of carbon in year $t+1$, r is the discount rate, g is the rate of growth of the social cost of carbon, and $x = r - g$ is the net carbon discount rate. The rental valuation measures the annualised benefit of temporarily storing one metric tonne of CO₂ for one year. The country-level storage value is the product of the in-country carbon stock, this rental price and a natural contribution ratio of 1,

$$GEP_{\text{coastal carbon}} = C_{\text{total}} \times \text{rental SCC} \times 1.$$

Because our base year is 2019, computing the one-year rental rate requires $P_C(2019)$ and $P_C(2020)$. The Greenhouse gas Impact Value Estimator (GIVE) model’s damage functions begin in 2020, making direct computation of $P_C(2019)$ infeasible. We use the GIVE model because it represents the current state of the art in SCC estimation, integrating four sectoral damage channels (mortality, agriculture, energy and sea level rise) with probabilistic socioeconomic projections from the Resources for the Future Socioeconomic Pathways (RFF-SPs), and forms the basis of the updated U.S. EPA social cost of carbon (Rennert et al. 2022). We therefore approximate $P_C(2019)$ using $P_C(2020)$. This approximation is justified on two grounds. First, the year-over-year growth rate of the SCC is modest: our replication shows approximately 1.6% growth from 2020 to 2021 (from \$185 to \$188 per tCO₂ at the 2.0% Ramsey rate), driven primarily by GDP growth and cumulative emissions. Second, the COVID-19 pandemic caused a sharp contraction in global GDP in 2020 which would have compressed the difference between the 2019 and 2020 SCCs. Together, the small baseline growth rate and the recessionary shock in 2020 imply that $P_C(2019)$ and $P_C(2020)$ are nearly identical.

Under this approximation, we adopt a carbon price of \$678.33 per metric tonne of carbon (tC), corresponding to the mean SCC estimated by Rennert et al. (Rennert et al. 2022) at a 2% discount

rate, converted from CO₂ to carbon units ($\$185.05/\text{tCO}_2 \times 3.664 \text{ tCO}_2/\text{tC}$). The resulting annual rental rate is approximately \$13.43 per tC in 2020 US dollars. To express this in constant 2019 dollars we adjust for inflation using the GDP deflator reported by the World Bank as 1.85% for 2019, giving $\$13.43 / 1.0185 = \13.19 per tC in 2019 US dollars. The same rental price is applied to mangrove, salt marsh and seagrass stocks, treating each tonne of carbon as fungible across ecosystems.

Across the four candidate Ramsey discount rates considered here (1.5%, 2.0%, 2.5%, 3.0%), the inflation-adjusted rental SCC in 2019 dollars ranges from \$8.51 per tC at a 3.0% discount rate to \$16.50 per tC at a 1.5% discount rate, with the 2.0% baseline used in this study sitting near the middle of this range at \$13.19 per tC. Table 1 summarises the rental price at these four rates; the 2.0% row is the baseline used in this study.

Table 1: Rental social cost of carbon at alternative Ramsey discount rates.

Discount rate	Growth rate (carbon price)	Net discount rate (discount rate – growth rate)	SC-CO (2020\$)	SCC (2020\$)	Rental SCC (2020\$)	Rental SCC (2019\$)
0.015	0	0.015	308	1,129.33	16.81	16.50
0.020	0	0.020	185	678.33	13.43	13.19
0.025	0	0.025	118	432.67	10.68	10.49
0.030	0	0.030	80	293.33	8.67	8.51

3 Results

Coastal blue carbon storage generates a gross economic product of approximately \$80.02 billion (2019\$) in the base year. The account covers 63.4 million hectares of mapped mangrove, salt marsh and seagrass habitat inside national EEZs, storing 6.07 Pg C valued at the baseline rental SCC of \$13.19 per tC.

Seagrass dominates the global value because the mapped seagrass extent is large and the 1 m soil-carbon constant applies across that extent. Mangroves have the highest per-hectare stocks but a smaller EEZ-area footprint in the final country account, while salt marshes contribute a mid-sized share from temperate and high-latitude coastlines.

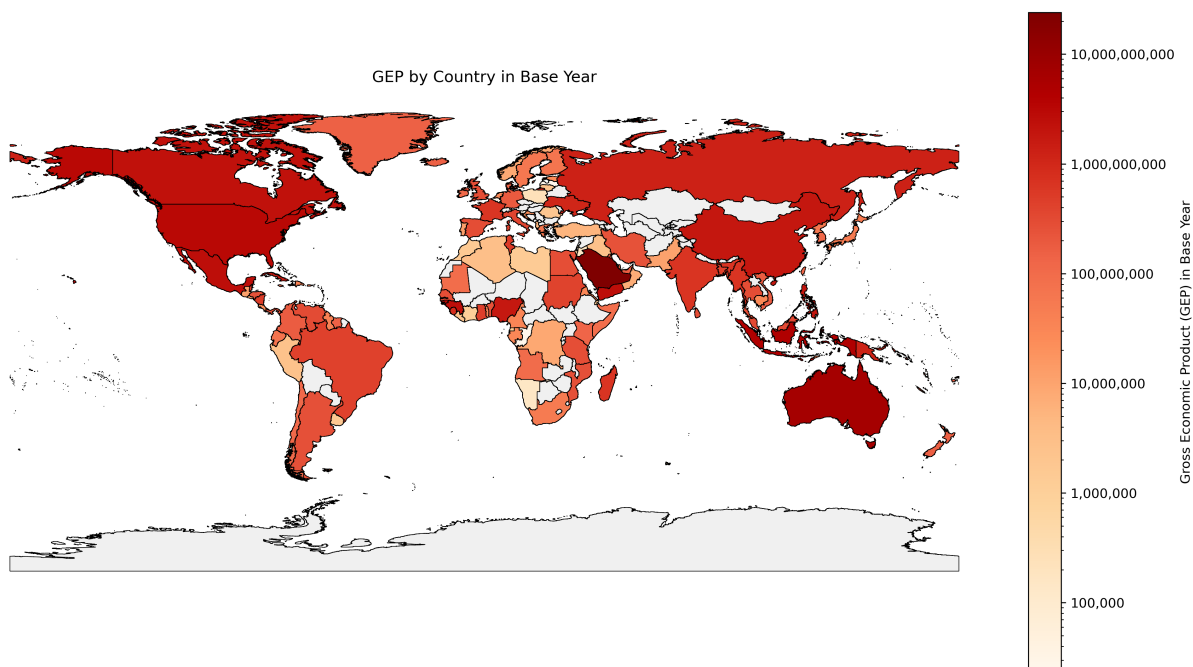
Table 2: Coastal carbon storage area, carbon stock and annual rental value by ecosystem in the base year.

	Ecosystem	Area (Mha)	Stock (Pg C)	Value (bn 2019\$)	Share of value (%)
0	Mangrove	0.37	0.66	8.72	10.90
1	Salt marsh	0.86	0.91	12.00	15.00
2	Seagrass	62.21	4.50	59.31	74.10

Because the country-level storage value scales linearly with the rental price, the global blue carbon storage value implied by the discount-rate range in Table 1 is approximately \$51.63 billion at a 3.0% rate and \$100.10 billion at a 1.5% rate.

The map below shows per-country storage value on a logarithmic colour scale. Countries with extensive seagrass or salt-marsh shelves rank highest in this baseline account, with Saudi Arabia, Australia and Indonesia forming the top three country totals.

Figure 2: Coastal carbon storage value by country in the base year.



Notes: Country values are shown on a logarithmic colour scale. Countries with no positive mapped coastal carbon storage value are shown in grey.

4 Validation

We validate the coastal-carbon results in two steps. First, we compare the monetary value of the global account against Bertram et al. (Bertram et al. 2021), who provide the closest published global country-level economic assessment of blue carbon wealth. Second, we compare the physical carbon-stock totals produced by the pipeline against ecosystem-specific published estimates for mangroves, salt marshes and seagrasses.

Bertram et al. (Bertram et al. 2021) find that global blue carbon wealth generated by carbon sequestration in coastal blue carbon ecosystems amounts to $\text{US}\$190.7 \pm 29.5 \text{ billion yr}^{-1}$, based on a global mean SCC of $\text{US}\$640.3 \pm 4.93 \text{ tCO}_2^{-1}$. Our baseline estimate is $\text{US}\$80.02 \text{ billion yr}^{-1}$ in 2019 dollars, based on an SC- CO_2 of $\text{US}\$185.05 \text{ tCO}_2^{-1}$ and an annual rental SCC of $\text{US}\$13.19 \text{ tC}^{-1}$. The lower value is therefore expected: Bertram et al. apply a substantially higher full SCC to annual carbon sequestration, while this study focuses on the annual rental value of standing carbon

stock within each country’s EEZ. The two estimates therefore differ both in valuation parameter and in the physical quantity being valued: annual sequestration in Bertram et al., and stored carbon stock here.

After this economic validation, we compare the physical stock estimates by ecosystem. Differences reflect (i) the extent layer chosen (GMW v3 2019 for mangroves; pre-processed GLC_FCS30D 2019 for salt marshes; WCMC v7.1 compilation for seagrass), (ii) the soil-depth convention (top 1 m throughout, sometimes deeper in field studies) and (iii) the per-pixel density product (Sanderman 2018 raster for mangrove SOC; latitude-step functions for salt marsh SOC and Fourqurean-scaled constant for seagrass SOC).

4.1 Mangrove

The pipeline produces a global mangrove total of **11.75 Pg C** over the GMW v3 2019 extent (~14.7 Mha), decomposed in Table 3.

Table 3: Global mangrove carbon stock, this study vs published estimates.

Component	This study (Pg C)	Published (Pg C)	Source
Above-ground biomass C	2.81	1.5 – 3.0	Hamilton & Friess (Hamilton and Friess 2018); Hutchison (Hutchison et al. 2014); Simard (Simard et al. 2019)
Below-ground biomass C	1.15	0.7 – 1.5	IPCC Wetlands Supplement ratios (IPCC 2014)
Soil organic C, top 1 m	7.79	6.4	Sanderman (Sanderman et al. 2018)
Total ecosystem C	11.75	5.72 ± 0.10 (2020, 13.6 Mha) ; 8.05 (2000, 6.2 Mha)	Ju (Ju et al. 2025); Wang (Wang et al. 2025)

Our mangrove SOC total (7.79 Pg C) is roughly 20 percent above the Sanderman et al. global figure of 6.4 Pg C (Sanderman et al. 2018), because the all-touched rasterisation of GMW v3 onto the 10 arc-second grid retains thin coastal fringes that the original 30 m Sanderman product does not, expanding the integrated extent. Our above-ground biomass (2.81 Pg C) sits at the upper end of the published range and is consistent with the Hamilton and Friess latitude regression (Hamilton and Friess 2018) over the broader GMW v3 extent. Our total (11.75 Pg C) is therefore higher than Ju et al. (Ju et al. 2025) and Wang et al. (Wang et al. 2025), a difference that is driven primarily by extent (we use a recent and inclusive extent layer) and only secondarily by the density products.

4.2 Salt marsh

The pipeline produces a global salt marsh total of **4.04 Pg C** over the GLC_FCS30D 2019 extent, decomposed in Table 4.

Table 4: Global salt marsh carbon stock, this study vs published estimates.

Component	This study (Pg C)	Published (Pg C)	Source
Above-ground biomass C	0.038	0.03 – 0.05	Chmura (Chmura et al. 2003)
Below-ground biomass C	0.086	0.07 – 0.15	Howard manual ratios (Howard et al. 2014)
Soil organic C, top 1 m	3.92	1.4 – 2.1 (MarSOC) ; 0.4 (McLeod 2011 review)	Maxwell (Maxwell et al. 2024); McLeod (McLeod et al. 2011)
Total ecosystem C	4.04	1.5 – 2.1	Maxwell (Maxwell et al. 2024)

Our salt marsh SOC (3.92 Pg C) is roughly twice the Maxwell et al. MarSOC global estimate of 1.4 – 2.1 Pg C (Maxwell et al. 2024). The principal driver is the latitude step function used here, which assigns 180 – 350 Mg C/ha across latitude bands; Maxwell et al. report a global area-weighted mean of ~225 Mg C/ha and a more conservative global extent. Replacing the step function with the native MarSOC raster would bring our salt marsh total back toward the Maxwell range, and is the most consequential improvement remaining in the pipeline for this ecosystem.

4.3 Seagrass

The pipeline produces a global seagrass total of **4.82 Pg C** over the WCMC v7.1 extent (~16 – 27 Mha, McKenzie 2020 confidence range), decomposed in Table 5.

Table 5: Global seagrass carbon stock, this study vs published estimates.

Component	This study (Pg C)	Published (Pg C)	Source
Above-ground biomass C	0.046	0.007 – 0.012	Gomis (Gomis et al. 2025)
Below-ground biomass C	0.107	0.017 – 0.028	Gomis (Gomis et al. 2025)
Total biomass C	0.153	0.024 – 0.040	Gomis (Gomis et al. 2025)
Soil organic C, top 1 m	4.67	4.2 – 8.4	Fourqurean (Fourqurean et al. 2012); Macreadie (Macreadie et al. 2021)

Component	This study (Pg C)	Published (Pg C)	Source
Total ecosystem C	4.82	4.3 – 8.4	Fourqurean (Fourqurean et al. 2012)

Our seagrass biomass-only total (0.153 Pg C) is roughly four to six times the Gomis et al. 2025 range of 0.024 – 0.040 Pg C (Gomis et al. 2025) because (i) ~78 percent of WCMC polygons receive the Gomis global mean of 1.55 Mg C/ha when their genus is not identified, which is conservative for stocks but generous when applied over the full WCMC extent rather than the McKenzie high-confidence subset, and (ii) the WCMC v7.1 extent of ~27 Mha is at the upper end of the published range. Our soil total (4.67 Pg C) sits within the Fourqurean 2012 lower bound, because the 70 Mg C/ha top-1 m constant is itself a conservative scaling of the Fourqurean top-20 cm mean.

5 Conclusion

We provide a country-level economic assessment of coastal blue carbon storage across marine EEZs, combining recent extent and density products for mangroves, salt marshes and seagrasses with the rental social cost of carbon at a 2 percent Ramsey discount rate. The approach extends the country-level blue carbon natural-capital framework of Bertram et al. (Bertram et al. 2021) from the annual sequestration flow to the standing-stock storage component, and aligns with the IPCC Tier 1 conventions for wetland carbon accounting (IPCC 2014).

The methodology is deliberately conservative on three axes. Soil depth is fixed at 1 m, in line with the IPCC Tier 1 convention (IPCC 2014) and with recent global stock-mapping efforts (Sanderman et al. 2018; Ju et al. 2025; Wang et al. 2025), although mangrove peats often extend to 2 m or beyond (Donato et al. 2011). Mangrove above-ground biomass is taken from the Hamilton and Friess latitude regression (Hamilton and Friess 2018) rather than canopy-height or bioclimatic models (Hutchison et al. 2014; Simard et al. 2019), trading some local accuracy for global parsimony. Tropical-wet versus tropical-dry below-ground allocation is latitude-only when no precipitation raster is configured; supplying mean annual precipitation refines this and improves estimates in dry-tropical mangroves of Saudi Arabia, Senegal and northern Australia.

Seagrass storage remains the weakest of the three ecosystems. The biomass component is genus-aware, but 78 percent of WCMC polygons lack a genus identification and the soil component is a uniform global constant rather than a spatial product. Future work should ingest a gridded seagrass SOC layer, no published equivalent of Sanderman 2018 or Maxwell 2024 exists at present, and split the global mean across bioregions, where the Mediterranean averages roughly 50 percent above the global mean (Gomis et al. 2025).

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7 Appendix

7.1 Top countries

	Country	Storage value (2019\$)
191	Saudi Arabia	24,078,258,749.32
14	Australia	6,696,019,775.73
102	Indonesia	4,711,213,375.75
234	United States	3,214,800,466.14
87	Guinea-Bissau	2,824,974,962.92
246	Yemen	2,518,176,073.78
142	Mexico	2,510,808,382.30
84	Guinea	2,432,666,029.38
39	Canada	2,300,253,320.61
176	Philippines	2,041,345,615.04

7.2 Coastal Carbon GEP by income group

	income_grp	value
0	1. High income: OECD	15,093,938,470.53
1	2. High income: nonOECD	25,809,536,966.97
2	3. Upper middle income	11,313,759,851.45
3	4. Lower middle income	18,506,655,192.54
4	5. Low income	9,296,835,372.37

7.3 Coastal Carbon GEP by UN region

	region_un	value
0	Africa	12,087,240,231.80
1	Americas	13,912,140,145.86
2	Antarctica	0.00
3	Asia	39,539,197,684.31
4	Europe	4,826,418,579.60
5	Oceania	9,615,612,230.66
6	Seven seas (open ocean)	40,116,981.61

7.4 Coastal Carbon GEP by continent

	continent	value
0	Africa	12,074,600,494.91
1	Antarctica	0.00
2	Asia	39,315,898,488.78
3	Europe	4,826,418,579.60
4	North America	12,304,142,692.74
5	Oceania	9,615,612,230.66
6	Seven seas (open ocean)	236,329,664.91
7	South America	1,647,723,702.24

7.5 Coastal Carbon GEP by subregion

	subregion	value
0	Antarctica	0.00
1	Australia and New Zealand	6,860,319,023.80
2	Caribbean	2,488,770,344.39
3	Central America	4,105,772,726.79
4	Central Asia	0.00
5	Eastern Africa	1,571,675,094.53
6	Eastern Asia	2,311,302,998.63
7	Eastern Europe	2,307,996,655.41

	subregion	value
8	Melanesia	1,911,266,265.97
9	Micronesia	315,973,566.67
10	Middle Africa	172,026,603.44
11	Northern Africa	1,221,669,738.29
12	Northern America	5,669,873,372.45
13	Northern Europe	865,469,103.30
14	Polynesia	528,053,374.23
15	Seven seas (open ocean)	40,116,981.61
16	South America	1,647,723,702.24
17	South-Eastern Asia	8,315,440,663.95
18	Southern Africa	53,132,243.11
19	Southern Asia	1,846,785,196.55
20	Southern Europe	793,499,063.10
21	Western Africa	9,068,736,552.43
22	Western Asia	27,065,668,825.18
23	Western Europe	859,453,757.79

7.6 Coastal Carbon GEP by Country in Base Year

	iso3_r250_name	value
0	Aruba	3,485,530.89
1	Afghanistan	0.00
2	Angola	98,880,156.17
3	Anguilla	832,388.84
4	Åland	0.00
5	Albania	4,996,366.06
6	Andorra	0.00
7	United Arab Emirates	170,591,568.66
8	Argentina	269,327,054.48
9	Armenia	0.00
10	American Samoa	40,993,317.84
11	Antarctica	0.00
12	French Southern Territories	390,732.50
13	Antigua and Barbuda	24,093,724.12
14	Australia	6,696,019,775.73
15	Austria	0.00
16	Azerbaijan	0.00
17	Burundi	0.00
18	Belgium	28,180.16
19	Benin	239,989,562.21
20	Bonaire, Sint Eustatius and Saba	0.00
21	Burkina Faso	0.00
22	Bangladesh	417,964,030.66
23	Bulgaria	0.00

	iso3_r250_name	value
24	Bahrain	40,504,236.58
25	Bahamas	387,409,316.56
26	Bosnia and Herzegovina	0.00
27	Saint Barthélemy	0.00
28	Belarus	0.00
29	Belize	519,214,397.55
30	Bermuda	33,645.53
31	Bolivia	0.00
32	Brazil	437,729,566.15
33	Barbados	6,768,259.71
34	Brunei	8,566,396.36
35	Bhutan	0.00
36	Bouvet Island	0.00
37	Botswana	0.00
38	Central African Republic	0.00
39	Canada	2,300,253,320.61
40	Cocos Islands	0.00
41	Switzerland	0.00
42	Chile	109,391,448.07
43	China	1,895,658,299.62
44	Côte d'Ivoire	1,133,764.59
45	Cameroon	33,568,550.34
46	Democratic Republic of the Congo	8,875,689.24
47	Republic of the Congo	23,569.19
48	Cook Islands	168,460.68
49	Colombia	180,940,223.87
50	Comoros	115,636,415.07
51	Cabo Verde	0.00
52	Costa Rica	23,084,557.04
53	Cuba	1,703,304,064.55
54	Curaçao	0.00
55	Christmas Island	0.00
56	Cayman Islands	4,127,283.52
57	Cyprus	6,462,477.38
58	Czechia	0.00
59	Germany	189,180,474.41
60	Djibouti	1,392,058.79
61	Dominica	44,139,142.71
62	Denmark	139,668,027.96
63	Dominican Republic	37,881,719.02
64	Algeria	3,204,109.34
65	Ecuador	72,341,045.18
66	Egypt	274,154,697.68
67	Eritrea	29,655,263.78
68	Western Sahara	0.00
69	Spain	299,894,138.35

	iso3_r250_name	value
70	Estonia	24,005,359.39
71	Ethiopia	0.00
72	Finland	67,653,669.71
73	Fiji	166,627,127.64
74	Falkland Islands	75,930,473.18
75	France	641,144,138.49
76	Faroe Islands	1,806,782.10
77	Micronesia	156,819,042.96
78	Gabon	29,604,361.62
79	United Kingdom	290,014,269.23
80	Georgia	0.00
81	Guernsey	2,167,881.65
82	Ghana	480,863,638.94
83	Gibraltar	0.00
84	Guinea	2,432,666,029.38
85	Guadeloupe	0.00
86	Gambia	28,441,909.83
87	Guinea-Bissau	2,824,974,962.92
88	Equatorial Guinea	1,074,276.87
89	Greece	56,939,126.36
90	Grenada	1,447,895.83
91	Greenland	154,201,702.06
92	Guatemala	11,524,520.20
93	French Guiana	0.00
94	Guam	40,069,658.67
95	Guyana	38,897,808.08
96	China, Hong Kong Special Administrative Region	66,016,072.97
97	Heard Island and McDonald Island	0.00
98	Honduras	252,916,862.52
99	Croatia	23,547,483.84
100	Haiti	0.00
101	Hungary	0.00
102	Indonesia	4,711,213,375.75
103	Isle of Man	0.00
104	India	653,123,133.19
105	British Indian Ocean Territory	0.00
106	Ireland	102,574,144.22
107	Iran	246,968,626.43
108	Iraq	2,870,738.69
109	Iceland	156,259,917.73
110	Israel	607,271.88
111	Italy	387,012,566.59
112	Jamaica	77,495,663.49
113	Jersey	9,460,796.66
114	Jordan	149,261.95
115	Japan	52,895,267.02

	iso3_r250_name	value
116	Kazakhstan	0.00
117	Kenya	127,809,361.12
118	Kyrgyzstan	0.00
119	Cambodia	29,088,019.95
120	Kiribati	0.00
121	Saint Kitts and Nevis	5,087,515.70
122	South Korea	119,922,760.45
123	Kuwait	138,284,149.52
124	Laos	0.00
125	Lebanon	0.00
126	Liberia	2,292,211.16
127	Libya	1,386,128.06
128	Saint Lucia	883,256.32
129	Liechtenstein	0.00
130	Sri Lanka	294,708,296.72
131	Lesotho	0.00
132	Lithuania	1,900,433.65
133	Luxembourg	0.00
134	Latvia	24,145.50
135	China, Macao Special Administrative Region	0.00
136	Saint-Martin	5,825,438.89
137	Morocco	1,481,076.13
138	Monaco	49,607.64
139	Moldova	0.00
140	Madagascar	655,135,300.24
141	Maldives	223,299,195.53
142	Mexico	2,510,808,382.30
143	Marshall Islands	44,569,679.82
144	North Macedonia	0.00
145	Mali	0.00
146	Malta	9,281,381.15
147	Myanmar	663,090,151.38
148	Montenegro	0.00
149	Mongolia	0.00
150	Northern Mariana Islands	0.00
151	Mozambique	262,886,551.73
152	Mauritania	96,605,351.61
153	Montserrat	1,185,593.43
154	Martinique	0.00
155	Mauritius	11,171,031.25
156	Malawi	0.00
157	Malaysia	326,724,046.76
158	Mayotte	0.00
159	Namibia	152,972.67
160	New Caledonia	161,360,143.04
161	Niger	0.00

	iso3_r250_name	value
162	Norfolk Island	0.00
163	Nigeria	1,789,965,997.98
164	Nicaragua	512,761,797.89
165	Niue	0.00
166	Netherlands	29,051,357.09
167	Norway	8,437,349.44
168	Nepal	0.00
169	Nauru	0.00
170	New Zealand	164,299,248.07
171	Oman	6,962,547.71
172	Pakistan	10,721,914.01
173	Panama	265,774,610.09
174	Pitcairn Islands	0.00
175	Peru	2,626,349.94
176	Philippines	2,041,345,615.04
177	Palau	74,515,185.22
178	Papua New Guinea	1,184,786,821.44
179	Poland	248,754.65
180	Puerto Rico	68,566,251.76
181	North Korea	59,279,916.44
182	Portugal	11,796,417.05
183	Paraguay	0.00
184	Palestine	0.00
185	French Polynesia	61,079,320.45
186	Qatar	97,552,614.81
187	Réunion	0.00
188	Romania	1,911,442.84
189	Russia	1,238,548,325.83
190	Rwanda	0.00
191	Saudi Arabia	24,078,258,749.32
192	Sudan	405,718,624.58
193	Senegal	275,605,134.19
194	Singapore	21,016,239.76
195	South Georgia and the South Sand	0.00
196	Saint Helena, Ascension and Tris	0.00
197	Svalbard and Jan Mayen	0.00
198	Solomon Islands	293,773,779.74
199	Sierra Leone	861,619,697.27
200	El Salvador	9,687,599.20
201	San Marino	0.00
202	Somalia	88,128,719.26
203	Saint Pierre and Miquelon	584,238.10
204	Serbia	0.00
205	South Sudan	0.00
206	São Tomé and Príncipe	0.00
207	Suriname	145,730,319.36

	iso3_r250_name	value
208	Slovakia	0.00
209	Slovenia	31,583.70
210	Sweden	61,496,326.06
211	Swaziland	0.00
212	Sint Maarten	4,202,969.49
213	Seychelles	1,468,705.63
214	Syria	0.00
215	Turks and Caicos Islands	101,215,343.45
216	Chad	0.00
217	Togo	34,578,292.36
218	Thailand	299,322,045.86
219	Tajikistan	0.00
220	Tokelau	0.00
221	Turkmenistan	0.00
222	Timor-Leste	35,992,567.54
223	Tonga	337,464,049.22
224	Trinidad and Tobago	6,579,909.63
225	Tunisia	535,725,102.50
226	Turkey	5,249,134.89
227	Tuvalu	66,038.13
228	Taiwan	117,530,682.13
229	Tanzania	278,391,687.66
230	Uganda	0.00
231	Ukraine	1,067,288,132.09
232	United States Minor Outlying Isl	39,726,249.11
233	Uruguay	1,750,295.52
234	United States	3,214,800,466.14
235	Uzbekistan	0.00
236	Vatican City	0.00
237	Saint Vincent and the Grenadines	464,989.63
238	Venezuela	313,059,118.40
239	British Virgin Islands	3,774,086.86
240	Virgin Islands, U.S.	0.00
241	Vietnam	179,082,205.55
242	Vanuatu	104,718,394.11
243	Wallis and Futuna	518,728.20
244	Samoa	87,763,459.70
245	Serbia	0.00
246	Yemen	2,518,176,073.78
247	South Africa	52,979,270.44
248	Zambia	0.00
249	Zimbabwe	0.00